

Automata Theory and Formal Languages CS 131

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Grading System

Quizzes	30
Midterm	30
Final	30
HW, Attendance, etc.	10
Total	100

Objectives

At the end of this topic, the students can

- Define Sets and discuss its operation
- Explain what is relations and function
- Discuss graphs and trees.
- Discuss the concepts of language and automata

Why study theory?

- Theory provides concepts and principles that help us understand the general nature of the Computer Science discipline.
- The field of computer science includes a wide range of special topics, from machine design to programming.
- Each model in automata theory plays important roles in several applied areas. Finite automata are used in text processing, compilers, and digital(hardware) design. Context-free grammar (CFGs) are used in programming languages and artificial intelligence.

An **automaton(plural, automata)** is a construct that possesses all the indispensable features of a digital computer. It accepts input, produces output, may have some temporary storage, and can make decisions in transforming the input into the output.

A **formal language** is an abstraction of the general characteristics of programming languages. A formal language consists of a set of symbols and some rules of formation by which these symbols can be combined into entities called sentences. A formal language is the set of all sentences permitted by the rules of formation.

Mathematical Reviews

Sets

A **set** is a (finite or infinite) collection of elements, without any structure other than membership.

To indicate that x is an element of the set S , we write

$$x \in S, \quad \text{e.g. } 1 \in \{0, 1, 2\}.$$

The statement that x is not in S is written as

$$x \notin S, \quad \text{e.g. } 3 \notin \{1, 0, 2\}.$$

A set is said to be **finite** if it contains a finite number of elements; otherwise it is **infinite**.

A set can be specified by enclosing some description of its elements in curly braces;

Example of Finite sets

$\{0,1,2\}$ – set of integers 0,1,2

$\{1,2,0\}$ - , the same set of integers 0, 1, 2

$\{\}$ or $\emptyset$ - empty set

$\{a, b, \dots, z\}$ - stands for all the lowercase letters of the English alphabet

Example of Infinite sets

$\{0,1,2,3,\dots\}$ – set of natural numbers

$\{2, 4, 6,\dots\}$ denotes the set of all positive even integers.

We can also use explicit notation, in which we write

$S=\{i:i>0, i \text{ is even}\}$ or $S=\{i|i>0, i \text{ is even}\}$

We read this as “ S is the set of all i , such that i is greater than zero, and i even,” implying, of course, that i is an integer.

Set operations

union ($\cup$) : $S_1 \cup S_2 = \{ x: x \in S_1 \text{ or } x \in S_2 \}$

Example $\{0\} \cup \{1,2\} = \{0,1,2\}$

$\{0,1\} \cup \{1,2\} = \{0,1,2\}$

$\{i: i > 0, i \text{ is even}\} \cup \{i: i > 0, i \text{ is odd}\} = \{1,2,3,\dots\}$

intersection ($\cap$): $S_1 \cap S_2 = \{ x: x \in S_1 \text{ and } x \in S_2 \}$

Example $\{0,1,2\} \cap \{1,2\} = \{1,2\}$

$\{0,1\} \cap \{1,2\} = \{1\}$

$\{i: i > 0, i \text{ is even}\} \cap \{i: i > 0, i \text{ is odd}\} = \emptyset$

difference $(-)$: $S_1 - S_2 = \{ x : x \in S_1 \text{ and } x \notin S_2 \}$

Example

$$\{0,1,2\} - \{1,2\} = \{0\}$$

$$\{0,1,2\} - \{2,3\} = \{0,1\}$$

$$\{i : i > 0, i \text{ is even}\} - \{i : i > 0, i \text{ is odd}\} = \{i : i > 0, i \text{ is even}\}$$

Another basic operation is **complementation**. The complement of a set S , denoted by S^c , consists of all elements not in S . To make this meaningful, we need to know what the **universal set** U of all possible elements is.

Example Set of all integer numbers

Set of all computer programs

Set of all possible grades for the course

Set of all lower case letter of the alphabets

If U is specified, then

$$\overline{S} = \{x : x \in U, x \notin S\}$$

the complement of S includes all elements of U that are not in S .

Example

- U is a set of all final grades, from 0 to 100.

$$S = \{x : x \geq 50\} \quad - \text{ passing grades}$$

$$\overline{S} = \{x : x < 50\} \quad - \text{ failing grades}$$

- U is a set of integers

$$S = \{i : i \text{ is even}\}$$

$$\overline{S} = \{i : i \text{ is odd}\}$$

The set with no elements, called the **empty set** or the **null set**, is denoted by $\emptyset$.

From the definition of a set, it is obvious that

$$S \cup \emptyset = S - \emptyset = S$$

$$S \cap \emptyset = \emptyset$$

$$\overline{\emptyset} = U$$

$$\overline{\overline{S}} = S$$

The following useful identities, known as **DeMorgan's laws**,

$$\overline{S_1 \cup S_2} = \overline{S_1} \cap \overline{S_2},$$

$$\overline{S_1 \cap S_2} = \overline{S_1} \cup \overline{S_2},$$

are needed on several occasions.

Size of a set -The size of a finite set is the number of elements in it; this is denoted by $|S|$.

Example $S = \{0,1,2\}$

$$|S| = 3$$

$$|\emptyset| = 0$$

A set $S1$ is said to be a **subset** of S if every element of $S1$ is also an element of S . We write this as

$$S1 \subseteq S,$$

Example

$$\{0,1\} \subseteq \{0,1,2\}$$

$$\{i:i>0 \text{ and } i \text{ is even}\} \subseteq \{0,1,2,3,\dots\}$$

If $S1 \subseteq S$, but S contains an element not in $S1$, we say that $S1$ is a **proper subset** of S ; we write this as

$$S1 \subset S.$$

If S_1 and S_2 have no common element, that is,

$S_1 \cap S_2 = \emptyset$, then the sets are said to be **disjoint**.

A given set normally has many subsets.

The set of all subsets of a set S is called the **powerset** of S and is denoted by 2^S . Observe that 2^S is a set of sets.

Example

If S is the set $\{a, b, c\}$, then its powerset is

$$2^S = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}.$$

$$2^{\{1,2\}} = \{\emptyset, \{1\}, \{2\}, \{1,2\}\}$$

$$2^\emptyset = \{\emptyset\}$$

If $|S| = n$ then $|2^S| = 2^n$. This is an instance of a general result; if S is finite, then

$$|2^S| = 2^{|S|}.$$

Cartesian product

For the Cartesian product of two sets, which itself is a set of ordered pairs, we write

$$S = S_1 \times S_2 = \{(x, y) : x \in S_1, y \in S_2\}.$$

Example Let $S_1 = \{2, 4\}$ and $S_2 = \{2, 3, 5, 6\}$.

$$S_1 \times S_2 = \{(2, 2), (2, 3), (2, 5), (2, 6), (4, 2), (4, 3), (4, 5), (4, 6)\}.$$

Note that the order in which the elements of a pair are written matters. The pair $(4, 2)$ is in $S_1 \times S_2$, but $(2, 4)$ is not.

The notation is extended in an obvious fashion to the Cartesian product of more than two sets; generally

$$S_1 \times S_2 \times \cdots \times S_n = \{(x_1, x_2, \dots, x_n) : x_i \in S_i\}.$$

Functions and Relations

A **function** is a rule that assigns to elements of one set a unique element of another set. If f denotes a function, then the first set is called the **domain** of f , and the second set is its **range**. We write

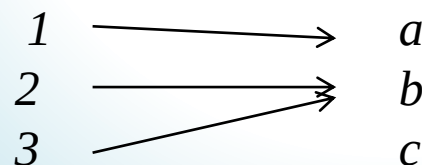
$$f: S_1 \rightarrow S_2$$

to indicate that the domain of f is a subset of S_1 and that the range of f is a subset of S_2 . If the domain of f is all of S_1 , we say that f is a **total function** on S_1 ; otherwise f is said to be a **partial function**.

Example

S_1

S_2



domain: {1,2,3}

range : {a,b}

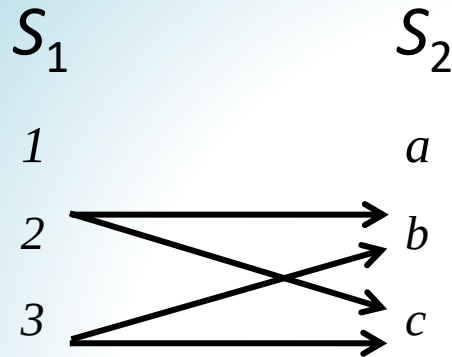
$$F(x) = x^2$$

domain: all real numbers

range : all non-negative real numbers

Relation - for each element of S_1 a relation determines several (possibly none) elements in S_2

Example



Example

Mapping from students classes they takes – a student can take none, one, or more classes.

Equivalence

A special case of a relation, usually denoted by “ $\equiv$ ”, which satisfies three properties;

1. $x \equiv x$ (reflexivity)
2. if $x \equiv y$, then $y \equiv x$ (symmetry)
3. if $x \equiv y$ and $y \equiv z$, then $x \equiv z$ (transitivity)

Graphs and Trees

A graph is a construct consisting of two finite sets, the set

$$V = \{v_1, v_2, \dots, v_n\} \text{ of } \mathbf{vertices}$$

and the set

$$E = \{e_1, e_2, \dots, e_m\} \text{ of } \mathbf{edges}.$$

Each edge is a pair of vertices from V , for instance,

$$e_i = (v_j, v_k)$$

is an edge from v_j to v_k .

We say that the edge e_i is an outgoing edge for v_j and incoming edge for v_k .

Such a construct is actually a directed graph (digraph), since we associate a direction(from v_j to v_k) with each edge.

Graphs may be labeled, a label being a name or other information associated with parts of the graph. Both vertices and edges may be labeled.

Graphs are conveniently visualized by diagrams in which the vertices are represented as circles and the edges as lines with arrows connecting the vertices. The graph with vertices $\{v_1, v_2, v_3\}$ and edges $\{(v_1, v_3), (v_3, v_1), (v_3, v_2), (v_3, v_3)\}$ is depicted in Figure 1.1.

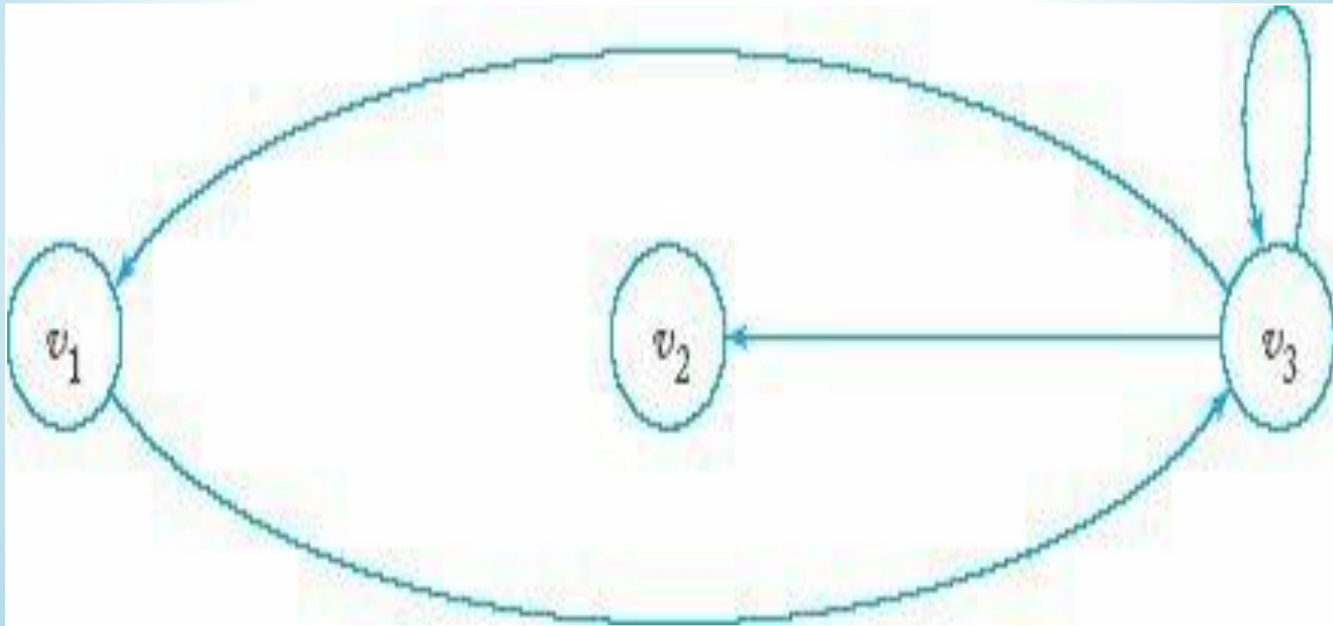


Figure 1.1 Graph

A sequence of edges $(v_i, v_j), (v_j, v_k), \dots, (v_m, v_n)$ is said to be a **walk** from v_i to v_n . The length of a walk is the total number of edges traversed in going from the initial vertex to the final one.

A **path** is a “walk” in which no edge is repeated.

A path is simple if no vertex is repeated.

A walk from v_i to itself with no repeated edges is called a **cycle** with base v_i .

In Figure 1.1 $(v_1, v_3), (v_3, v_2)$ is a simple path from v_1 to v_2 . The sequence of edges $(v_1, v_3), (v_3, v_3), (v_3, v_1)$ is a cycle, but not a simple one. If the edges of a graph are labeled, we can talk about the label of a walk. This label is the sequence of edge labels encountered when the path is traversed. Finally, an edge from a vertex to itself is called a **loop**. In Figure 1.1, there is a loop on vertex v_3 .

Trees are a particular type of graph. A tree is a directed graph that has no cycles, and that has one distinct vertex, called the **root**, such that there is exactly one path from the root to every other vertex.

This definition implies that the root has no incoming edges and that there are some vertices without outgoing edges. These are called the **leaves** of the tree. If there is an edge from v_i to v_j , then v_i is said to be the **parent** of v_j , and v_j the **child** of v_i . A node without children is called a **leaf**. The **level** associated with each vertex is the number of edges in the path from the root to the vertex. The **height** of the tree is the largest level number of any vertex. These terms are illustrated in Figure 1.2.

At times, we want to associate an ordering with the nodes at each level; in such cases we talk about **ordered trees**.

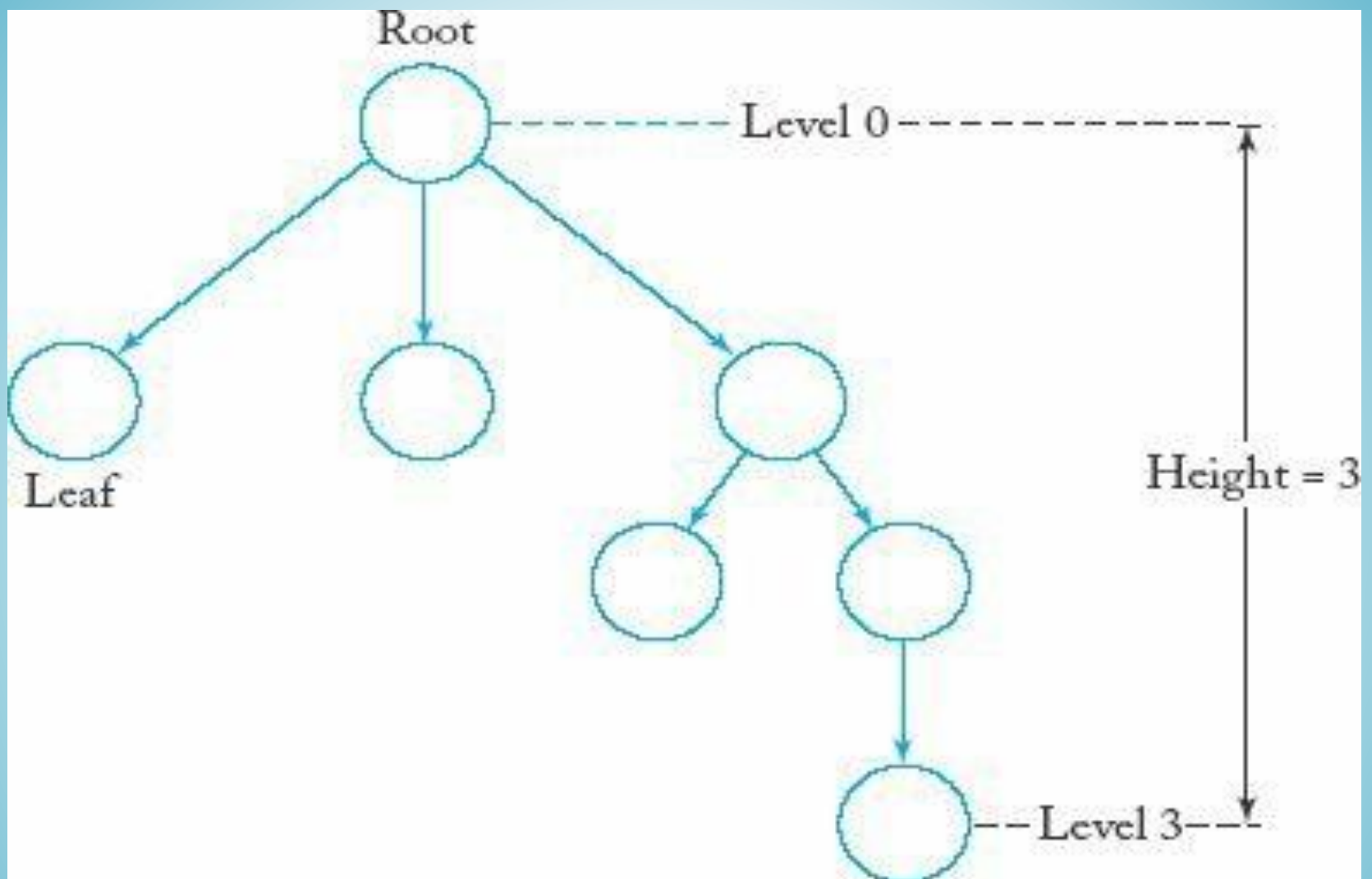


Figure 1.2 Tree

Activity

Answer the following:

1. $\{1,3,4, 5\} \cup \{ 1,2,6\}$
2. $\{a,b,c\} \cap \{a,b,e,f\}$
3. $\{11,23, 34\} \cup \{6, 11, 22\}$
4. $\{ 54, 12, 34,13\} - \{ 53, 12,14,13\}$
5. $\{0,1,c,d\} \cap \{a, b, e, f, 2, 3\}$
6. $\{ a, b, c, d, o \} - \{b, c, a, e, d\}$
7. $2^{\{0,1,2\}}$
8. $\{a,b,c\} \times \{ A,B\} \times \{0,1,3\}$
9. U is $\{ 0, 1, 3, 5, 7, 9, 11, 12, 14,15, 17, 18, 19\}$ and $S = \{x: x>6 \text{ and } x \text{ is even}\}$

What is the compliment of S?
10. What is the length of size of the compliment of S in problem 9?